

# RYAN LEITE ALBUQUERQUE

SENIOR SOFTWARE ENGINEER

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Senior backend engineer, 26 years in production. Shares ownership of a production ML platform for automated issue triage and duplicate detection – FastAPI over Kafka, Milvus vector search and Snowflake on Kubernetes/AWS EKS – and ships internal platforms solo, from data model to deploy: team onboarding (Django/GraphQL/Celery), an engineering-metrics platform with nine CI-fed dashboards, and AI developer tooling on the Claude Agent SDK and MCP. Production systems in Java, PHP, Ruby and Python. Earlier, led a 10-engineer team through a CMMI Level 3 appraisal as a trained SCAMPI evaluator. M.Sc. in Computer Science (Artificial Intelligence).

## EXPERIENCE

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### Principal Technical Manager • Backend & Platform Engineering

Jan 2026 – Present

CESAR (Centro de Estudos e Sistemas Avançados do Recife) • Recife, PE, Brazil

Backend and platform engineering for ML-driven developer-productivity systems: shared ownership of a production ML platform, four internal platforms and real-time apps built solo, AI/LLM developer tooling, and quality programs across five repositories. Kubernetes deployments authored as Helm charts, with Kustomize overlays.

- Shared ownership of a production ML platform for automated issue triage and duplicate detection: FastAPI, Kafka, Milvus vector search, Snowflake, Kubernetes on AWS EKS.
  - Developed a team onboarding platform (Django 6, GraphQL, Celery, Kubernetes) that automates a manual nine-step process, with progress tracking, user notifications and update triggers.
  - Developed an engineering-metrics platform (Django + GraphQL, Nuxt 4, PostgreSQL on Kubernetes) with nine dashboards over coverage, LOC, pull-request and bug-fix data from CI.
  - Added a self-service web app that triggers metric collection on demand with live log streaming, plus weekly Kubernetes CronJobs covering three repositories.
  - Shipped AI developer tooling: an autonomous Claude Agent SDK bug-fixer that opens pull requests behind a confidence gate, and an MCP server serving six triage tools to Claude Code, Cursor and Xcode.
  - Delivered two real-time team apps: planning poker on Django Channels WebSockets, and retrospectives on GraphQL with a six-state workflow engine.
  - Raised test coverage across five repositories – one from 0% to ~80% by authoring its entire test suite, the other four from roughly 50% to ~80% – and wired coverage reporting into CI.
  - Standardized the Python toolchain across the same five repositories: Poetry to uv, pre-commit to pre-commit, plus ruff and type-check gates.
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## Senior Technical Manager · Python Backend Services

Oct 2022 – Dec 2025

CESAR (Centro de Estudos e Sistemas Avançados do Recife) · Recife, PE, Brazil

Python backend services for two concurrent web platforms, standardized on JSON:API response contracts and backed by end-to-end CI/CD ownership with automated quality gates.

- Built and maintained backend services for two concurrent web platforms in Python (Django and Flask).
- Designed and implemented REST APIs to the JSON:API specification, standardizing response contracts across teams and reducing integration rework between backend and frontend squads.
- Owned CI/CD end to end for both platforms (Docker, Git/GitFlow, Jenkins) – automated build, test and deploy, with formatting, lint and test gates enforced in CI.
- Set up and ran Sentry error monitoring for both platforms – creating projects per environment and instrumenting the frontend and backend applications.

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## Senior Technical Manager · Full-Stack Delivery & Technical Leadership

Jun 2013 – Sep 2022

CESAR (Centro de Estudos e Sistemas Avançados do Recife) · Recife, PE, Brazil

Nine years across four client projects: four production web applications in four language stacks, the CI/CD environments behind them, and technical direction for the Scrum teams delivering them.

- Contributed to the development of four production web applications across nine years and four client engagements, each in a different language stack – PHP (Yii), Python (Flask/Falcon), Ruby on Rails and JavaScript (Backbone/Marionette, EmberJS, VueJS).
- Built and maintained CI/CD environments with Git/GitFlow, TeamCity, Jenkins, Ansible and HashiCorp Vault, enabling secure, repeatable and fully automated deployments.
- Served as technical lead across four client projects with Scrum teams of 6 to 15 engineers, running sprint planning, code review practice and delivery commitments.

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## Technical Manager · Enterprise Java & Mobile

Jan 2009 – Jun 2013

CESAR (Centro de Estudos e Sistemas Avançados do Recife) · Recife, PE, Brazil

An enterprise Java web and mobile application built end to end, the CI/CD and Scrum tooling behind it, and a nationwide production platform maintained against live traffic.

- Built an enterprise web and mobile application end to end using Java EE/ME, Android, Apache Struts 2, Apache CXF, Spring, Hibernate, Oracle and Apache Tomcat.
  - Implemented a complete CI/CD environment (Subversion/Git, Maven, TeamCity, Tomcat) for automated builds and deployments.
  - Built a Scrum task-board integration for Mantis (PHP) that gave distributed remote teams shared sprint visibility.
  - Maintained a nationwide production web platform while shipping new features against live traffic.
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## Senior Software Configuration Management Engineer

Jan 2005 - Jan 2009

CESAR (Centro de Estudos e Sistemas Avançados do Recife) · Recife, PE, Brazil

Led the company-wide Software Configuration Management team through a formal CMMI 1.2 Level 3 appraisal, defining the SCM processes, tooling and automation adopted as company standards.

- Led a 10-engineer Software Configuration Management team through a successful formal CMMI 1.2 Level 3 appraisal, serving as a trained SCAMPI evaluator.
  - Defined and maintained company-wide SCM processes as SCM lead within the Software Engineering Process Group (SEPG).
  - Rolled out SCM processes across multiple concurrent projects using Rational ClearCase, ClearDDTS, CVS, Subversion, Mantis, Apache Ant and Lintbuild.
  - Built automation tooling and Mantis extensions (PHP, Shell) for defect tracking, requirements traceability and code review reporting, later adopted as company-wide standards.
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## Software Configuration Management Engineer

Oct 2001 - Jan 2005

CESAR (Centro de Estudos e Sistemas Avançados do Recife) · Recife, PE, Brazil

SCM process definition and automation for multi-site mobile device programs, contributing to CMM Level 2 certification and to a ~20% rise in client evaluation scores.

- Defined and implemented SCM processes for multi-site mobile device development projects using Rational ClearCase and ClearDDTS, contributing to the organization's CMM Level 2 certification.
  - Raised client evaluation scores for SCM performance by approximately 20% by automating configuration management activities across roughly twelve projects.
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## Junior Software Engineer

Jan 2001 - Oct 2001

CESAR (Centro de Estudos e Sistemas Avançados do Recife) · Recife, PE, Brazil

The institute's first J2ME mobile development practice: 15+ applications shipped, including an internationally awarded game.

- Pioneered J2ME application development at CESAR, establishing the institute's first mobile development practice.
  - Developed 15+ mobile applications, primarily games, for early Java-enabled handsets.
  - Authored Sea Hunter, awarded Best Application from the Americas at the Asia Java Mobile Challenge.
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## Junior Software Engineer

Jun 2000 - Dec 2000

CESAR (Centro de Estudos e Sistemas Avançados do Recife) · Recife, PE, Brazil

Java web development: a J2SE WAP application for lottery game management, and an e-commerce portal supporting M.Sc. dissertation research.

- Developed a WAP application in J2SE for lottery game management.
- Built an e-commerce portal used to validate a team member's M.Sc. dissertation research.

SKILLS

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| <div>BACKENDExpert</div> <div>Python, Django, FastAPI, Flask, Falcon, GraphQL, Strawberry GraphQL, REST API, JSON:API, Celery, Kafka, WebSockets, PHP, Yii, Laravel, Ruby on Rails, Java</div> | <div>FRONTENDAdvanced</div> <div>JavaScript, TypeScript, VueJS, NuxtJS, PrimeVue, Tailwind CSS, Pinia, EmberJS, Backbone/Marionette, HTML, CSS, SCSS/SASS, BEM</div>                                                      |
| <div>AI &amp; ML ENGINEERINGAdvanced</div> <div>LLM application development, Model Context Protocol (MCP), Claude Agent SDK, Agentic tooling, Vector search, Milvus, Snowflake</div>           | <div>DEVOPS &amp; CI/CDExpert</div> <div>CI/CD, Jenkins, TeamCity, GitLab CI, Docker, Multi-arch builds, Ansible, HashiCorp Vault, Sealed Secrets, uv, mise, prek, ruff, pytest, Bash, Git, GitFlow, GitHub, GitLab</div> |
| <div>CLOUD &amp; ORCHESTRATIONAdvanced</div> <div>Kubernetes, Helm, Kustomize, AWS EKS, AWS, Google Cloud, mTLS, OAuth 2.0, Sentry, Error monitoring</div>                                     | <div>DATABASESAdvanced</div> <div>PostgreSQL, MySQL, Oracle, Redis, Milvus, Snowflake</div>                                                                                                                               |
| <div>PROCESS &amp; LEADERSHIPExpert</div> <div>Software Configuration Management, CMMI, SCAMPI appraisal, Scrum, Kanban, Technical Leadership, Code Review, Mentoring</div>                    |                                                                                                                                                                                                                           |

EDUCATION

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| <div>Federal University of Pernambuco (UFPE)</div> <div>Mar 2000 - Aug 2002</div> <div>M.Sc. in Computer Science</div> <div>Specialization in Artificial Intelligence. Dissertation: "KAGENT: A Java API for Intelligent Agents in Mobile Devices" (advisor: Geber Lisboa Ramalho; available at <a href="https://repositorio.ufpe.br/handle/123456789/2601">https://repositorio.ufpe.br/handle/123456789/2601</a>).</div> <div>Artificial Intelligence · Multi-Agent Systems · Software Engineering</div> |  |
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| <div>Federal University of Ceará (UFC)</div> <div>Jan 1996 - Dec 1999</div> <div>B.Sc. in Computer Science</div> |  |
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PUBLICATIONS

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| <div>Software Configuration Management Diversity: A Commercial Case Study</div> <div>2008-07</div> <div>BCS-CMSG – The CMDB and CMS: the Powerhouse of Service Management</div> <div>Second of two authors. Commercial case study on sustaining diverse software configuration management practices inside one organization, published while leading CESAR's company-wide SCM team.</div> |  |
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## **Desenvolvendo o Projeto Conceitual de Jogos Wireless (Developing the Conceptual Design for Wireless Games)** 2002-10

1st Brazilian Workshop in Games and Digital Entertainment (WJOGOS), SIBGRAPI 2002, Fortaleza, Brazil – pp. 141-144

Fifth of six authors. Conceptual design methods for wireless games, drawn from CESAR's early J2ME mobile game development.

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## **Embedding J2ME-based Inference Engine in Handheld Devices: The KEOPS Case Study** 2002

Proceedings of the First International Workshop on Ubiquitous Agents on Embedded, Wearable and Mobile Devices, Bologna, Italy

First author. Case study on embedding an inference engine in J2ME for resource-constrained handheld devices.

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## **KSACI: A Handheld Device Infrastructure for Agents Communication** 2002

Intelligent Agents VIII (LNAI 2333), ATAL 2001, Seattle, USA – Springer, pp. 423-435

First author. Communication infrastructure for agents running on handheld devices – built on the SACI framework, using KQML as the outer language and XML as the inner one, and demonstrated on early J2ME cell-phone emulators.

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### **AWARDS**

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## **Best Application from the Americas – Sea Hunter** 2002

Asia Java Mobile Challenge (SingTel, Optus and Sun Microsystems)

Sea Hunter, a J2ME mobile game written for early Java-enabled handsets at CESAR (Brazil), was named one of the Top 20 finalists selected from more than 1,000 entrants across 23 countries.

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### **CERTIFICATES**

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## **AI for Developers (54h)** 2025-08

CESAR School

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## **Certified ScrumMaster (CSM)** 2017-10

Scrum Alliance

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## **Zend PHP Certified Developer** 2014-10

Zend Technologies, Ltd.

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## **Rational ClearCase Fundamentals for Windows (16h)** 2002

Rational Software do Brasil

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Rational ClearCase Administration for Windows (16h)

Rational Software do Brasil

2002

Applying Rational ClearCase Metadata for Windows (8h)

Rational Software do Brasil

2002

LANGUAGES

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| <div><div>PORTUGUESE</div><div>Native speaker</div></div> | <div><div>ENGLISH</div><div>Professional working proficiency</div></div> |
|-----------------------------------------------------------|--------------------------------------------------------------------------|